

ELECTIVE COURSE-4.4

URA745: VIRTUAL REALITY				
	L	T	P	Cr
	2	2	0	3.0
Course Objectives: The objective of this course is to provide a comprehensive understanding of Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR), and Extended Reality (XR), including their concepts, market trends, and technologies; to develop AR/VR applications using game Engine; and to deploy XR applications across various platforms.				
Syllabus Introduction to AR/VR: Concepts and History: Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR), Extended Reality (XR); Market Analysis: Overview and current trends; Hardware and Software: Integrated Development Environments (IDEs), AR and VR device classification; Technologies: Non-Fungible Tokens (NFTs), Internet of Things (IoT), Artificial Intelligence (AI), Machine Learning (ML); Perception: Temporal resolution, spatial resolution, motion perception, depth perception, color perception, auditory perception, haptics perception, and locomotion interfaces. Unity3D: Editor: Overview and usage; Game Objects and Components, Materials and Texturing; Lighting: Basics and skybox, Package Management: Import and export; Modeling: Basics and Probuilder modeling, Terrain Creation; Asset Store: Introduction and Utilization, Avatar Creation and Animation: Animator and particle systems; Scripting: C# scripting; AR Development Platform: Vuforia with Unity. Introduction to Unreal Engine: Workflow: Basics of Unreal Engine workflow; Asset Libraries: Usage and management; Import Functions: Basics, Plug-in Connection, Blueprint Workflows, Lighting and Shading, Extended Reality and Augmented Reality Deployments, Extended Reality Deployment Platforms; Overview: wearable headset, STEAM, Open XR; Deployment Workflows: Introduction and Best Practices; Test Cases: sample programs from Volvo Group. AR-VR Industrial case-studies: Industry 4.0, Medical Robotics, Automobile Industry, Manufacturing Industry, Healthcare Industry. Extended Reality: XR Toolkit Setup, Assembly Smart Factory; Locomotion: Move and turn, Conveyor System Animation, Teleoperation, UI Interaction; XR Challenges: Best methods and processes, HMD Oculus, UX design, quality testing of XR systems.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Course Learning Outcomes (CLOs)

After completion of this course, students will be able to:

1. understand Core Concepts and Technologies in AR, VR, MR, and XR.
2. apply modeling, scripting, animation for AR/VR development using a game engine.
3. design and develop XR applications for deployment on AR/VR platforms.
4. create AR/VR modules for industrial applications addressing relevant challenges and adhering to best practices.

Text Books

1. Ralf Doerner, Wolfgang Broll, Paul Grimm, and Bernhard Jung, Virtual and Augmented Reality (VR/AR): Foundations and Methods of Extended Realities (XR), Springer, 2022.
2. Steve Aukstakalnis, Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR, Addison-Wesley, 2016.
3. Gregory Kipper and Joseph Rampolla, Augmented Reality: An Emerging Technologies Guide to AR, Elsevier, 2012.

Reference Books

1. Alan B. Craig: Understanding Augmented Reality: Concepts and Applications, Morgan Kaufmann, 2013.
2. Mark Billinghurst, Adrian Clark, and Gun Lee: A Survey of Augmented Reality, Foundations and Trends® in Human-Computer Interaction, 2015.
3. Jason Jerald: The VR Book: Human-Centered Design for Virtual Reality, ACM Books, 2015.
4. Tony Parisi: Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile, O'Reilly Media, 2015.
5. Jesse Glover: Unity AR & VR by Tutorials: Augmented and Virtual Reality Apps for iOS and Android, Razeware LLC, 2019.
6. Grigore C. Burdea and Philippe Coiffet: Virtual Reality Technology, Wiley-Interscience, 2003.
7. Christian Sandor and Steven Feiner: Handbook of Augmented Reality, Springer, 2011.
8. Tzu-Chieh (Zijay) Tang: Practical Augmented Reality: A Beginner's Guide to AR, VR, and MR, Apress, 2018.
9. Dietrich Thalmann: The Social Life of Avatars: Presence and Interaction in Shared Virtual Environments, Springer, 2001.

Evaluation Scheme:**Evaluation Scheme:**

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
	Evaluation Elements	Weightage (%)

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